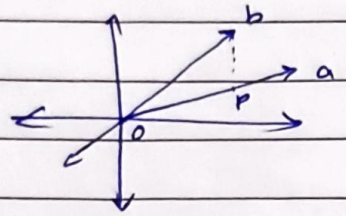
Projectile Calculation

Derivation



e = error

$e = b - p$

we know that p is a multiple of a

$p = xa$

since e is perpendicular to a

$$a^T e = 0$$
$$a^T(b - xa) = 0$$
$$a^T b - a^T x a = 0$$
$$a^T b = a^T x a$$
$$\frac{a^T b}{a^T a} = x$$

$$p = x\, a$$
$$p = \frac{a^T b}{a^T a}\, a$$

$$p = \frac{a a^T}{a^T a}\, b$$

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solve for $b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\quad a = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

$a = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ $\quad a^T = \begin{bmatrix} 4 & 0 \end{bmatrix}$

$a a^T = \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$

$PM = \frac{a a^T}{a^T a} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$a^T a = 16 + 0 = 16$

$p = PMb = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = p$

so the point of projection is

(20)

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